

Question

Sodium cyclopentadienyl dicarbonylferrate(0) reacts with 2-chloroethyl methyl sulfide to generate compound **A** with the molecular formula $C_{10}H_{12}FeO_2S$; **A** generates **B** upon heating. **B** exhibits two carbonyl stretching vibrational infrared absorption peaks. It is known that sodium cyclopentadienyl dicarbonylferrate(0) can be obtained by reacting an iron complex **X** that satisfies the 18-electron rule with sodium, and sodium cyclopentadienyl dicarbonylferrate(0) is the only product. The following statements are made:

1. The cyclopentadienyl dicarbonylferrate(0) anion carries 2 negative charges.
2. Iron in both **A** and **B** satisfies the 18-electron rule.
3. **B** has a bridging carbonyl structure.
4. **X** is a mononuclear complex.

What is the sum of the numbers of the correct statements?

A. 2

B. 5

C. 7

D. 9

E. 10

F. 11

G. 13

H. None of the above answers are correct.

Answer

Correct Answer: **A**

Detailed Explanation

Sodium cyclopentadienyl dicarbonyl iron(0) satisfies the 18-electron rule. The charge carried by the negative ion is $18 - 8 - 5 - 2 \times 2 = 1$. Statement 1 is incorrect.

CHECKPOINT

0.5 PTS

Cyclopentadienyl dicarbonyl iron(0) anion carries a 1- negative charge

A is the product of the S_N2 reaction of cyclopentadienyl dicarbonyl iron(0) anion with 2-chloroethyl methyl sulfide ($\text{ClCH}_2\text{CH}_2\text{SCH}_3$), i.e., $(\text{C}_5\text{H}_5)(\text{CO})_2\text{FeCH}_2\text{CH}_2\text{SCH}_3$. If sulfur does not coordinate to iron, then there are $8 + 5 + 2 \times 2 + 1 = 18$ electrons around iron, satisfying the 18-electron rule.

CHECKPOINT

0.5 PTS

Iron in **A** satisfies the 18-electron rule

B has two different carbon-oxygen bonds, and bridging carbonyls may exist. **B** is a dimer of **A**. However, **A** has no driving force for dimerization, so this reaction pathway is excluded. In addition, β -elimination is uncommon when iron is the central metal, so it is excluded. Another possibility is that the alkyl ligand of **A** migrates to CO to form an acyl ligand, and the product is $(\text{C}_5\text{H}_5)(\text{CO})\text{Fe}[\text{C}(\text{O})\text{CH}_2\text{CH}_2\text{SCH}_3]$. At this time, there are only 16 electrons around iron, and the sulfur atom of the acyl ligand can coordinate to iron, making iron satisfy the 18-electron rule, and the stability is improved. Statement 2 is correct, and statement 3 is incorrect.

CHECKPOINT

1 PTS

Alkyl ligand migrates to CO

CHECKPOINT

1 PTS

Sulfur atom of the acyl ligand coordinates to iron

CHECKPOINT

0.5 PTS

Iron in **B** satisfies the 18-electron rule

$\text{Na}[\text{Fe}(\text{C}_5\text{H}_5)(\text{CO})_2]$ is obtained by the reaction of sodium and **X**, and the product is unique, so the simplest formula of **X** is $[\text{Fe}(\text{C}_5\text{H}_5)(\text{CO})_2]$. At this time, there are $8 + 5 + 2 \times 2 = 17$ electrons around iron, which does not satisfy the 18-electron rule, so there is a metal-metal bond in **X**, which should be $[\text{Fe}(\text{C}_5\text{H}_5)(\text{CO})_2]_2$. Statement 4 is incorrect.

CHECKPOINT

0.5 PTS

X is $[\text{Fe}(\text{C}_5\text{H}_5)(\text{CO})_2]_2$

In summary, the correct statement is 2, choose option A.